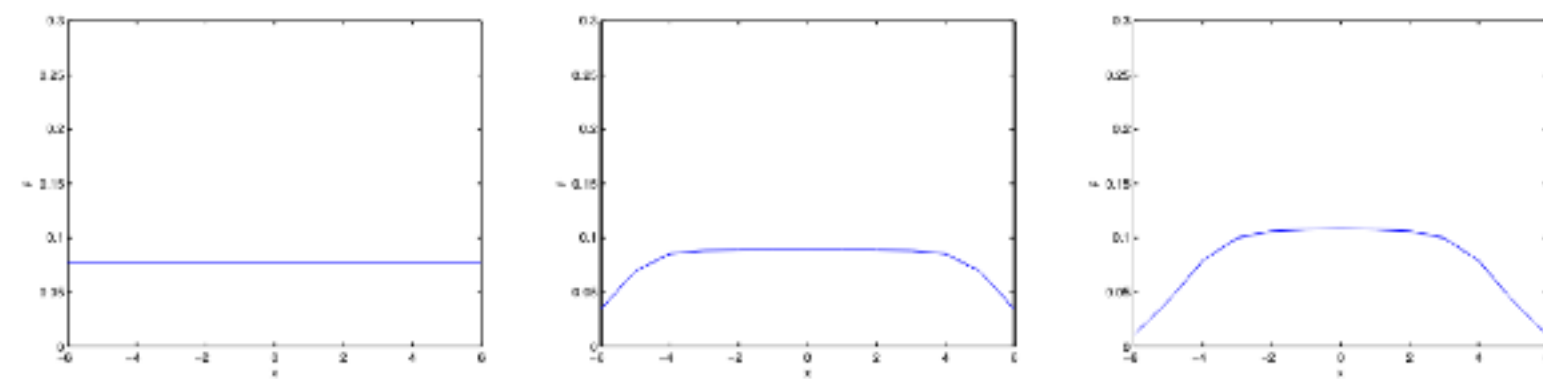


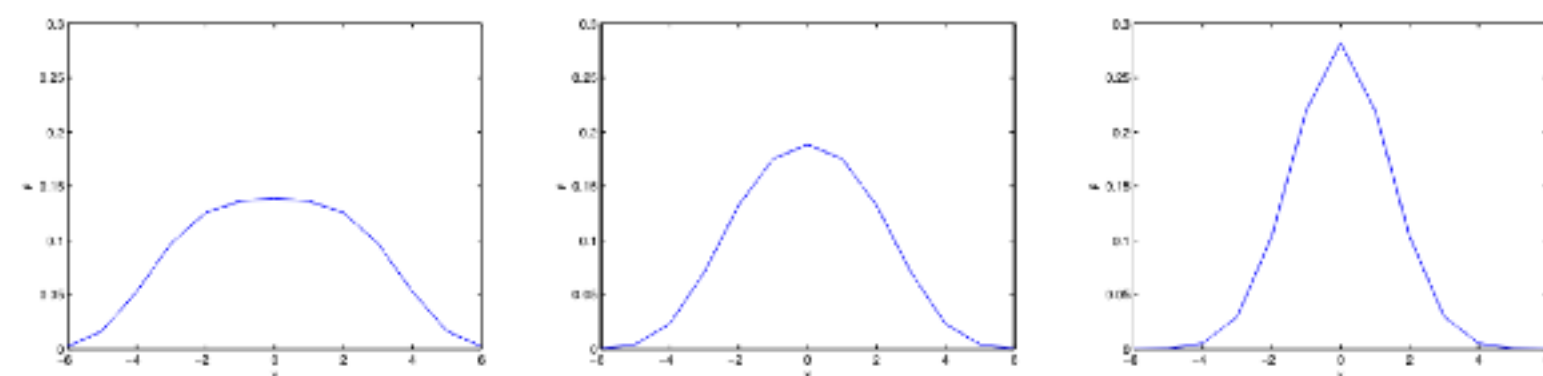
Related work: SBP or OT on graphs for discrete state space



(A) $t=0$

(B) $t=0.2$

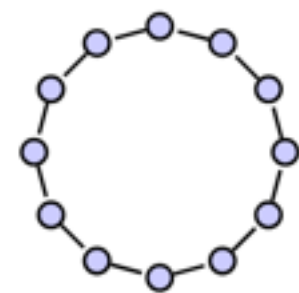
(C) $t=0.4$



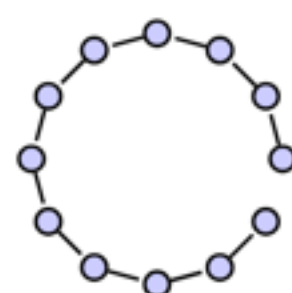
(D) $t=0.6$

(E) $t=0.8$

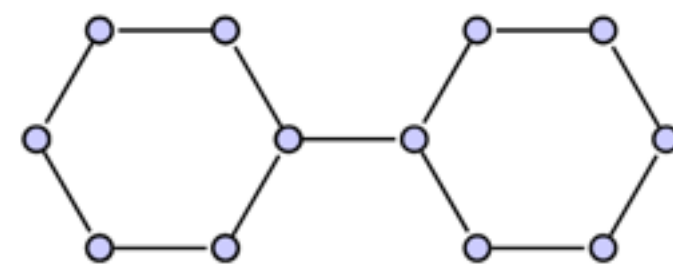
(F) $t=1$



29.2



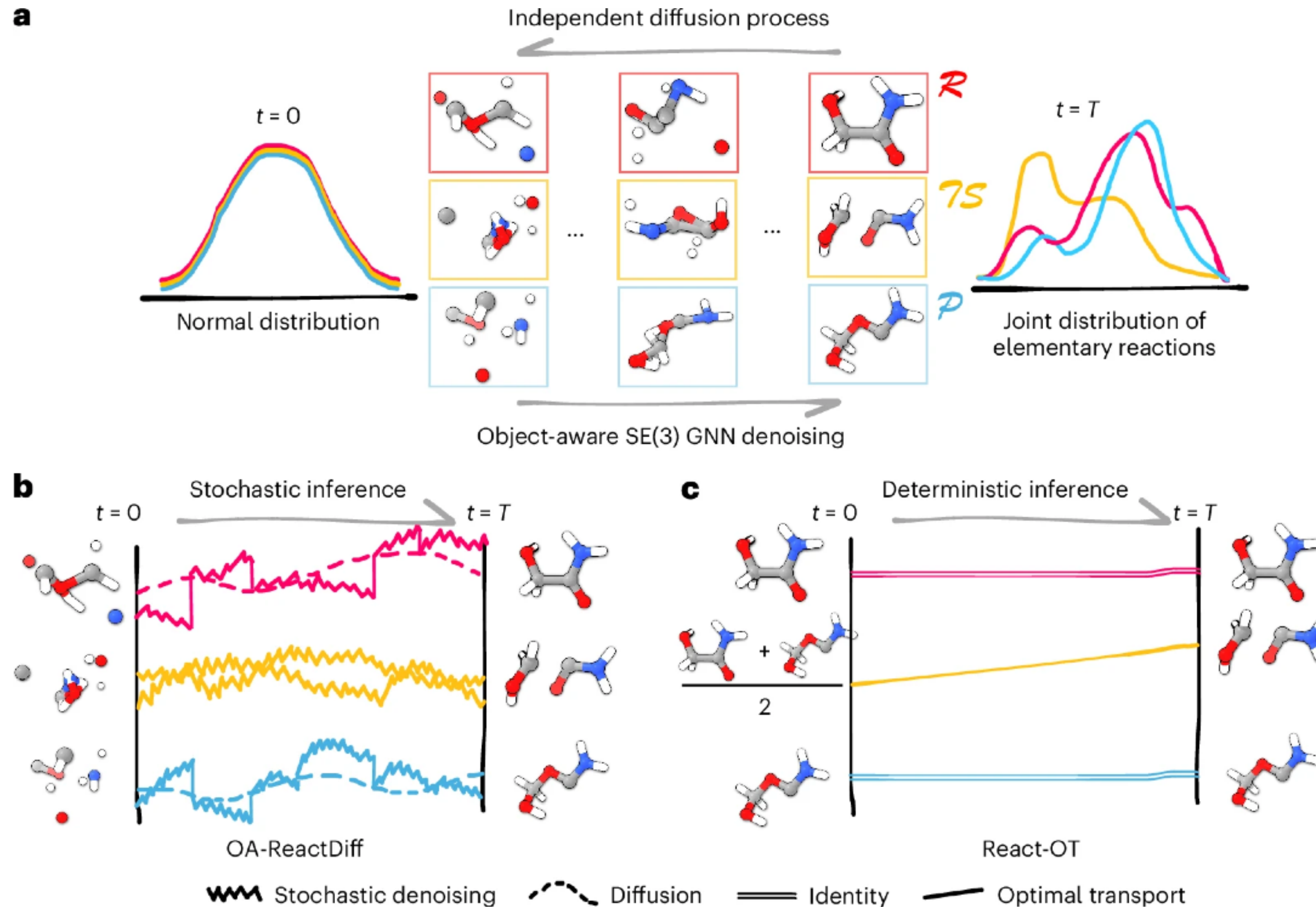
30.5



8.0

- Graphs as samples of discrete distributions on node state space $\sum p_i = 1$
- CTMC as the stochastic process
- SBP / OT on graphs, SOC form
$$\min_{p,b} \int_0^1 \frac{1}{2} \|b\|^2 dt \quad \text{s.t.} \quad b - \text{controlled Kolmogorov}$$
- Connected to Boltzmann distribution — adjoint sampling?
- Some works on this: [graph generation Xu et al. 2025](#)
- CTMC $\sim$ Hodge decomposition
- For both molecular generation, and protein optimization

Similar work 1 for generating transition states in chemical reactions [Duan et al. 2025 nature MI]



Here:

- positions modeled as distributions on graphs
- Forward: noising
- SE(3) GNN transition kernel (LEFTNet)

Our work:

- Topological stochastic heat diffusion

$$dY_t = -cLY_t dt + g_t dW_t$$

- Topology-aware transition kernels